UIC Department of Chemical Engineering
UNIVERSITY OF ILLINOIS AT CHICAGO
COLLEGE OF ENGINEERING

ASSIGNMENT #02

Course:	**CHE 433: PROCESS SIMULATION WITH ASPEN PLUS**
Term:	**Fall 2025**
Professor:	**Dr. Michael Caracotsios**
Office:	**EIB 244**
Cell Phone:	**(630) 696-0934**
E-mail:	**mcaracot@uic.edu**

Student names: Aaron Casas, Qussay Suleiman

Due Date: **Friday October 10th, my Mailbox (3:00PM)**

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a typed report of your assignment.
- Include this cover page with your name(s) in the report.
- Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
- Bring a hard copy of your report as a PDF file by the Due Date and Time in class

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

Problem A: Optimal Cooling Water Flow Rate in Condenser

Your engineering team has been asked to design a single-pass condenser for the condensation of propane vapor at 20 bar pressure. The vapor is at its dew point, and the flow rate is 2088 kmol/hr. Cooling water at 25 °C is available from a nearby cooling tower. Estimate the optimal water circulation rate in kg/s.

Data:

Heat Exchanger Type: Shell and Tube Single Pass

Purchase cost of the Heat Exchanger ($ January 2010): $C_{PURCHASE} = 28,000 + 54.0\,A^{1.2}$ $\quad A[=]m^2$

Heat Transfer Area: A

Lang Factor: $LF = 3.5$

Annualized Capital Charge Ratio: $ACCR = 0.2$

Annualized Capital Cost ($/year): $ACC = C_{PURCHASE} * LF * ACCR$

Cost of Cooling Water: $C_{CW} = \$14.8 / 1000 m^3$

Overall Heat Transfer Coefficient: $U = 425.6\ W / \left(m^2 \cdot K\right)$

Heat Capacity of Water: $C_P = 4.184\ kJ / \left(kg \cdot K\right)$

Operating Year: $8000\,hrs$

Cost Escalation Factor: 1.50

Section #1: PFD

The single-pass condenser was modelled using a FAKE heat exchanger, which revolves around using two heater units in parallel, rather than a single heat exchanger. This was done mainly to simplify the design specifications of the unit. A hierarchy using a real heat exchanger was used to import the heat exchanger area when calculating capital costs. Both the diagram for the FAKE and REAL exchanger are shown in figure 1 and figure 2, respectively.

Figure 1: Main Process Flow Diagram

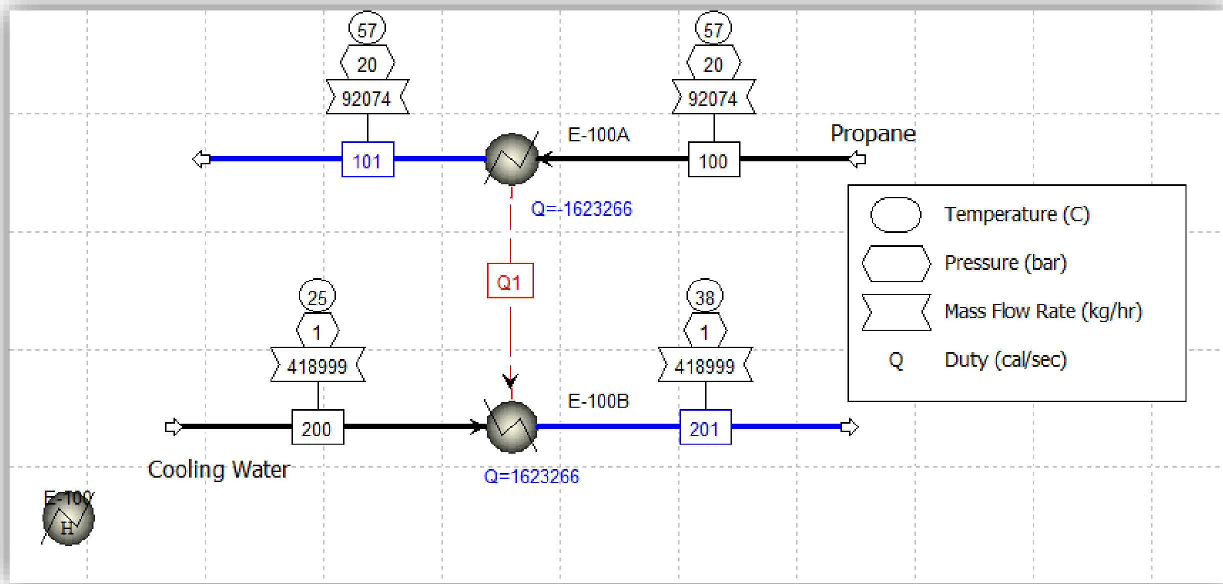
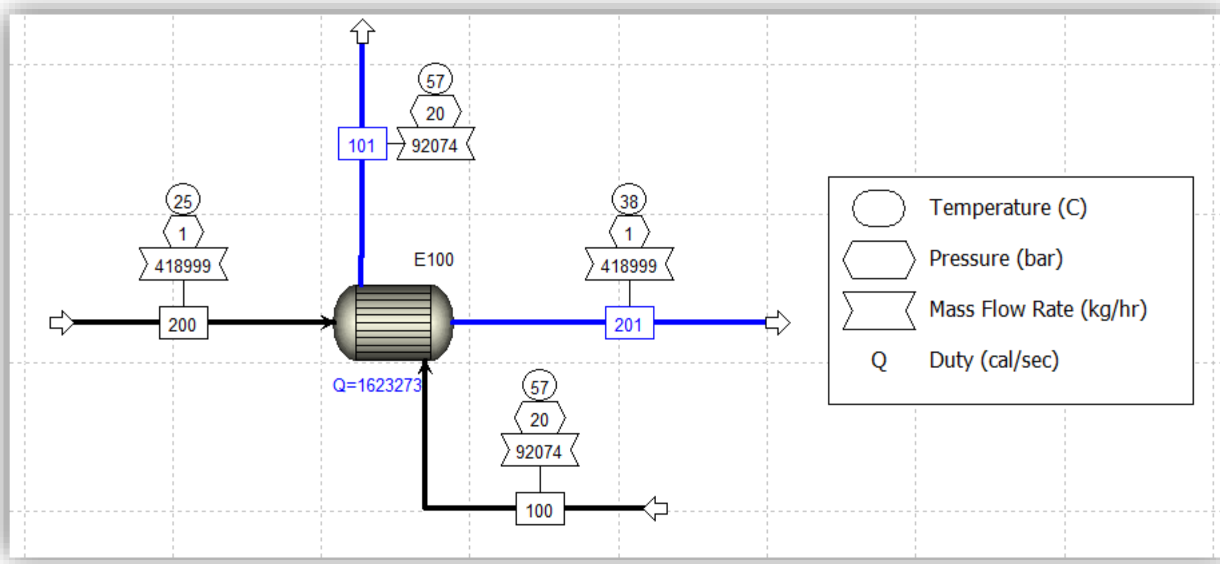


Figure 2: REAL HEX Process Flow Diagram



Section #2: Heat and Weight Balance

Table 1 presents the stream summary output from Aspen Plus V14 for the condensation system. The data were exported to Excel and reformatted for clarity and consistency with the UIC template. Four unused rows (Description, MIXED Substream, Mass Fractions, and Mole Fractions) were removed since they contained no data.

The system was modeled using two heaters: one for the propane stream (Streams 100–101) and one for the water stream (Streams 200–201). A heat exchanger was later included in a hierarchy for cost estimation, which produced similar results as the two-heater configuration.

Table 1: Stream Summary

Stream Property	Units	100	101	200	201
Phase		Vapor Phase	Liquid Phase	Liquid Phase	Liquid Phase
Temperature	C	57.0047	56.8406	25.0000	37.9116
Pressure	bar	20.0000	19.9311	1.0000	0.9311
Molar Vapor Fraction		1.0000	0.0000	0.0000	0.0000
Mass Flows	kg/sec	25.5760	25.5760	116.3886	116.3886
Mole Flows	kmol/hr	2088.0000	2088.0000	23257.9813	23257.9813
Molar Enthalpy	cal/mol	-25024.6376	-27823.3720	-68725.8549	-68474.5967
Molar Entropy	cal/mol-K	-69.6776	-78.1542	-40.1199	-39.2948
Enthalpy Flow	cal/sec	-14514289.7859	-16137555.7676	-444006845.6562	-442383579.6745
Average MW		44.0965	44.0965	18.0153	18.0153
Molar Density	mol/cc	0.0011	0.0099	0.0552	0.0545
Heat capacity, mixture	cal/mol-K	25.0472	37.8253	19.4663	19.4619
Availability (flow), mixture	cal/sec	-2465143.3381	-2622585.1611	-366727455.8248	-366693512.7795
Exergy flow rate	cal/sec	919350.9473	761909.1244	-43.2960	33899.7483

Section #3: Solution Logic and description

The solution logic for this condensation system was developed using Aspen Plus V14. The process was modeled with two heaters representing the hot and cold sides of the heat exchanger, and a separate exchanger was later introduced within a hierarchy solely for cost estimation purposes. A calculator block was used to link the heat exchanger area and water mass flow rate to a Fortran-based cost model, while an optimizer was implemented to minimize the total cost of operation.

The flowsheeting structure consists of two heaters arranged in parallel. The propane heater represents the hot side, where the inlet stream was specified at 20 bar and a vapor fraction of 1.0 to simulate propane entering at its dew point. The outlet vapor fraction was fixed at 0.0 to condense the propane completely, and a small pressure drop is simulated by setting the pressure to -1 psi. The water heater represents the cold side, with an inlet temperature of 25 °C and a pressure of 1 bar. A similar pressure drop was applied across this heater. The heat stream travels from the hot side (propane) to the cold side (water). These two heaters simulate the thermal behavior of the real exchanger, while the exchanger block in the hierarchy mirrors their duties and is used to extract the heat transfer area for cost analysis.

Two Transfer blocks (one for the hot side and one for the cold side) copy the relevant inlet stream states from the two-heater model to the inlet streams of the real heat exchanger in the hierarchy, ensuring the exchanger is sized under the same conditions. In that exchanger, a pressure drop is set, the hot-outlet vapor fraction is fixed at 0.0 and the overall U is specified; the area computed there is exported.

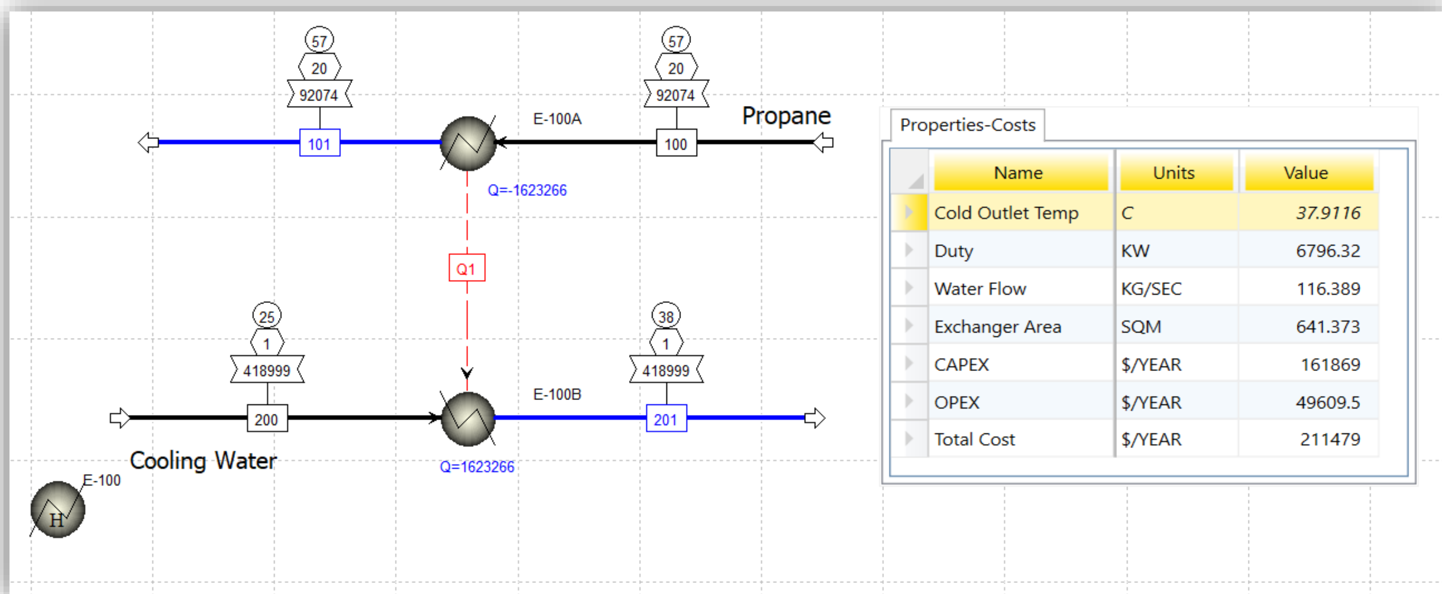
A calculator block was implemented to compute the capital, operating, and total costs of the heat exchanger. The purchase cost was determined using the provided purchase cost equation as a function of area, and the capital cost was then obtained by applying the Lang factor, annual capital charge rate (ACCR), and cost escalation factor. The operating cost was derived by calculating the annual cooling water volume from the water mass flow rate and density, then multiplying by the unit cost of $14.8 per 1000 m^3. The total cost was found by summing the CAPEX and OPEX. Among the imported variables, only the water mass flow rate and heat exchanger area were directly used in the Fortran code, while other variables (such as duty) were transferred for result display.

An optimization block was used to minimize the total cost by varying the inlet cooling water mass flow rate. A constraint was applied to keep the outlet temperature of the cooling water below 45 °C, ensuring that the actual temperature rise remained within the expected range of approximately 15 °C. It was observed that the optimizer's final result varied slightly (by 1–2%) depending on the initial mass flow rate guess specified in the inlet stream; guesses closer to the optimal value yielded results more quickly and with higher accuracy.

Section #4: Miscellaneous results

While no specific KPIs were required, several key thermodynamic and cost parameters were compiled into a custom table for clarity and documentation. After executing the optimization routine, the optimal cooling-water circulation rate was found to be approximately **116 kg/s**, corresponding to a temperature rise of about 13 °C across the exchanger. The results show that the capital cost ($161,869/year) is significantly higher than the operating cost ($49,609/year), which is expected since the exchanger's large surface area and construction dominate initial expenses, while utility costs remain relatively low. The combined total annual cost was calculated to be $211,479.

Figure 3: Significant Properties and Costs custom table



Name	Units	Value
Cold Outlet Temp	C	37.9116
Duty	KW	6796.32
Water Flow	KG/SEC	116.389
Exchanger Area	SQM	641.373
CAPEX	$/YEAR	161869
OPEX	$/YEAR	49609.5
Total Cost	$/YEAR	211479

Section 5: References

Note that since no additional discussion was requested for this assignment, the references section replaces the discussion section.

This problem was relatively straightforward, and all information used to solve it was derived from the materials provided in the problem description or from lessons taught by Dr. Michael Caracotsios. No external sources or research articles were consulted, however, the cost of equipment used information from Gavin's textbook. Aspen Plus V14 was used to perform the simulation and costing calculations, while ChatGPT was utilized to assist with navigating Aspen and refining the clarity and formatting of this report.

References

Aspen Technology, Inc. (2024). *Aspen Plus V14* [Computer software]. AspenTech. https://www.aspentech.com/

OpenAI. (2025). *ChatGPT (GPT-5)* [Large language model]. OpenAI. https://chat.openai.com/

Towler, Gavin, and Ray Sinnott. *Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design*. 3rd ed. Butterworth-Heinemann, 2021.

Problem B: Optimization of a refrigeration loop

Methane at temperature 25 °C and pressure 30 bar is to be precooled down to -38 °C prior to liquefaction. The precooling will be done using propane as the refrigerant. Design a multi-stage refrigeration loop and calculate the annual operating cost and the capital cost. You may attempt to minimize the total annualized cost for your design. The flow rate of methane is 250 MMSCFD. The isentropic efficiency of the compressors can be taken as 80% and the mechanical efficiency equal to 95%.

Notes:

For both problems assume that the physical properties of methane and propane are best described by the Peng-Robinson Equation of State

Minimum temperature approach for condenser may be taken as $\Delta T_{\min} = 10\,^{o}C$

Minimum temperature approach for evaporators:

Stage	$\Delta T_{\min}$
1	8
2	6
3	4
4	2

Gavin Towler and Ray Sinnott, **Chemical Engineering Design**. Principles Practice and Economics of Plant and Process Design, Second Edition © Elsevier Ltd. 2013.

Data:

Heat Exchangers: Shell and Tube Single Pass

Purchase cost of the Heat Exchanger ($ January 2010): $C_{PURCHASE} = 28,000 + 54\,A^{1.2}$ $\quad A[=]m^2$

Heat Transfer Area: A

Lang Factor: $LF = 3.5$

Compressors: Centrifugal

Purchase cost of the Compressor ($ January 2010): $C_{PURCHASE} = 580,000 + 20,000\,P^{0.6}$ $\quad P[=]kW$

Power Requirement: P

Lang Factor: $LF = 2.5$

Valves: Expansion Valves

Purchase cost of the Valve ($ January 2010): $C_{PURCHASE} = 8,142\,m^{0.7}$ $\quad m[=]kg/s$

Mass Flow Rate: m

Lang Factor: $LF = 1.3$

Drums: Flash Drum with Hemispherical Head and Bottom

Purchase cost of the Drum ($ January 2010): $C_{PURCHASE} = 11,600 + 34W^{0.85}$ $W[=]kg$

Shell Mass: W

Lang Factor: $LF = 4.0$

Data (Continued):

Annualized Capital Charge Ratio: $ACCR = 0.2$

Annualized Capital Cost ($/year): $ACC = C_{PURCHASE} * LF * ACCR$

Cost of Cooling Water: $C_{CW} = \$0.354 / GJ$

Cost of Electricity: $C_{EL} = \$0.06 / kWh$

Overall Heat Transfer Coefficient Condenser: $U = 425.6\ W / \left(m^2 \cdot K\right)$

Overall Heat Transfer Coefficient Evaporators: $U = 425.6\ W / \left(m^2 \cdot K\right)$

Heat Capacity of Water: $C_P = 4.184\ kJ / \left(kg \cdot K\right)$

Cooling Water Temperature: $T_{CWin} = 25\,^{o}C$

Cooling Water Pressure: $P_{CWin} = 20\ psia$

Cooling Water Temperature Rise: $10\,^{o}C$

Operating Year: $8000\,hrs$

Cost Escalation Factor: 1.50

Additional Instructions for your Report:

Your Custom table should contain three tabs

Tab 1 (Name KPIs): Display COP, Total Power(kW), Total Annualized Capital Cost (MM$/oper_year), Total Utility Cost (MM$/oper_year) and the sum of the two costs

Tab2 (Name: Detailed Cost): The installed cost ($) of each equipment and cost of each utility in $/oper_year

Tab3 (Name: Optimal Pressure): The final discharge pressure of each stage in bar

Section 1: PFD

Out of all the different compressor stages for the refrigerant cycle, we found that a 4-stage compressor is the most cost-efficient. The cost was determined using the constants and equations provided in this assignment, with the fake units and the hierarchy set up to match the real heat exchanger unit.

Figure 1: Process Flow Diagram

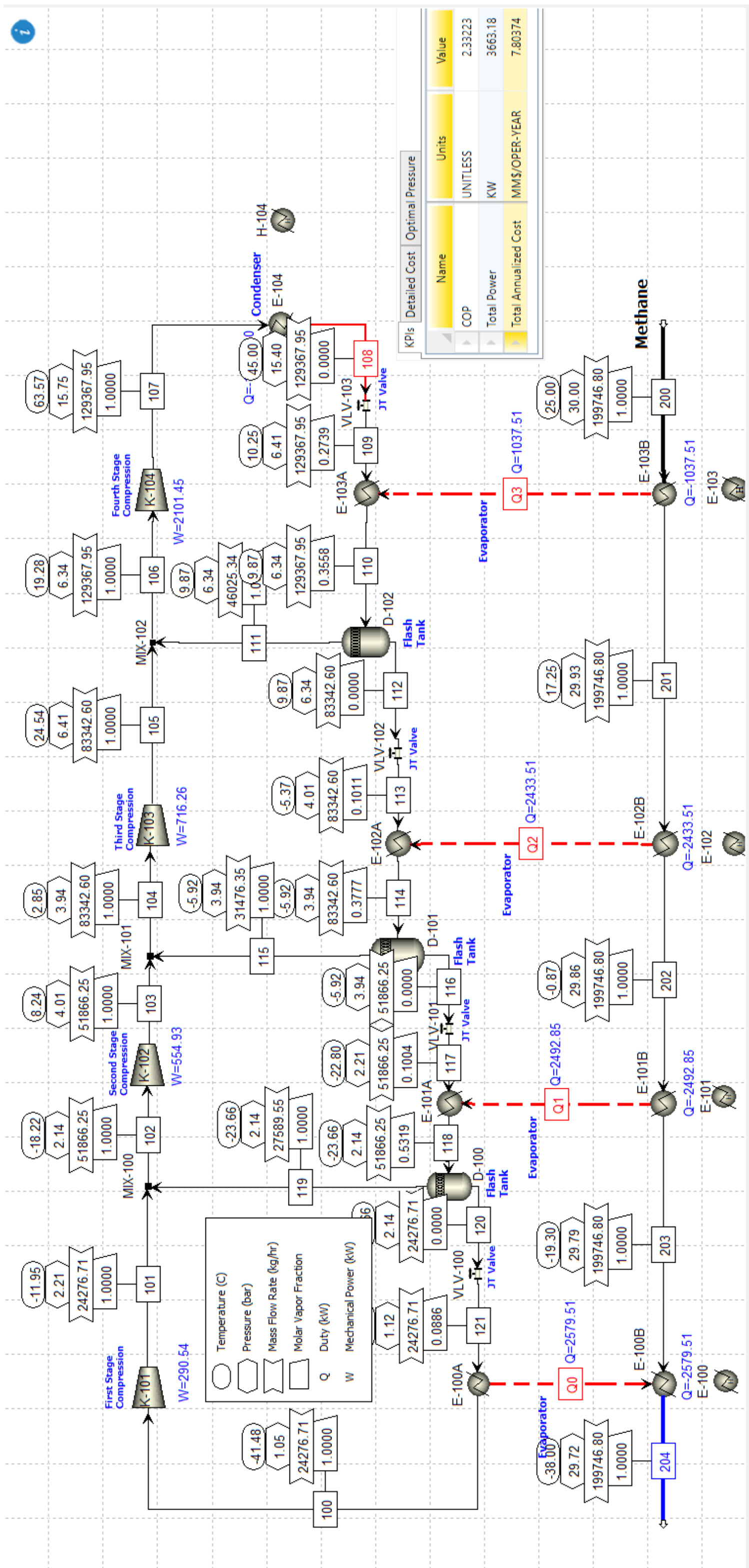


Figure 2: Evaporator Hierarchy

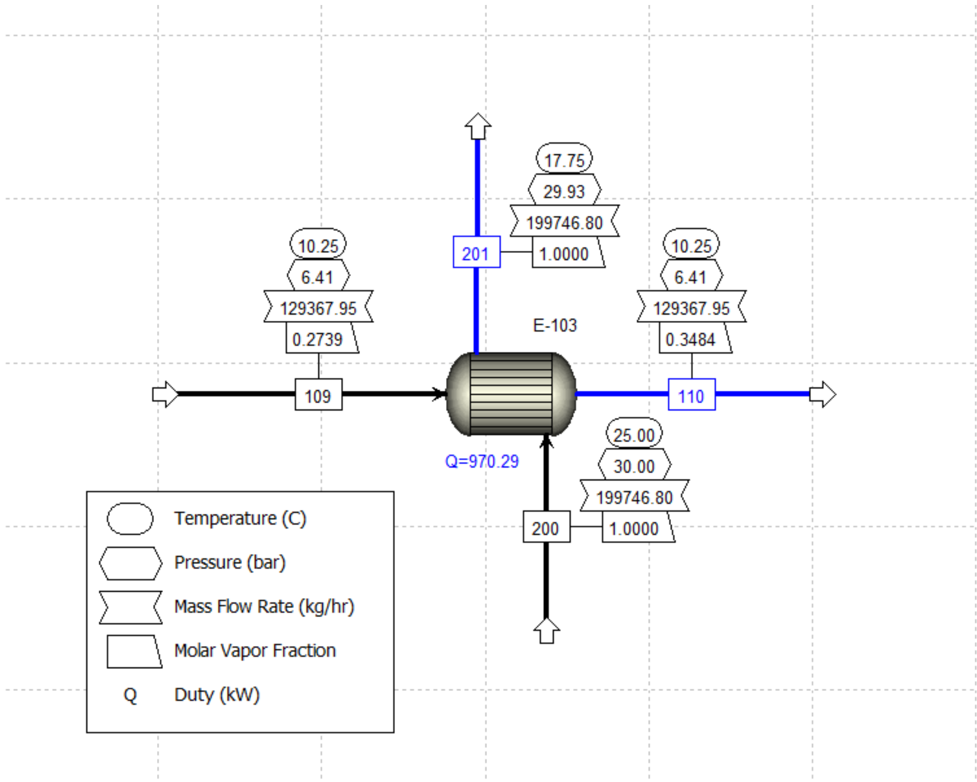
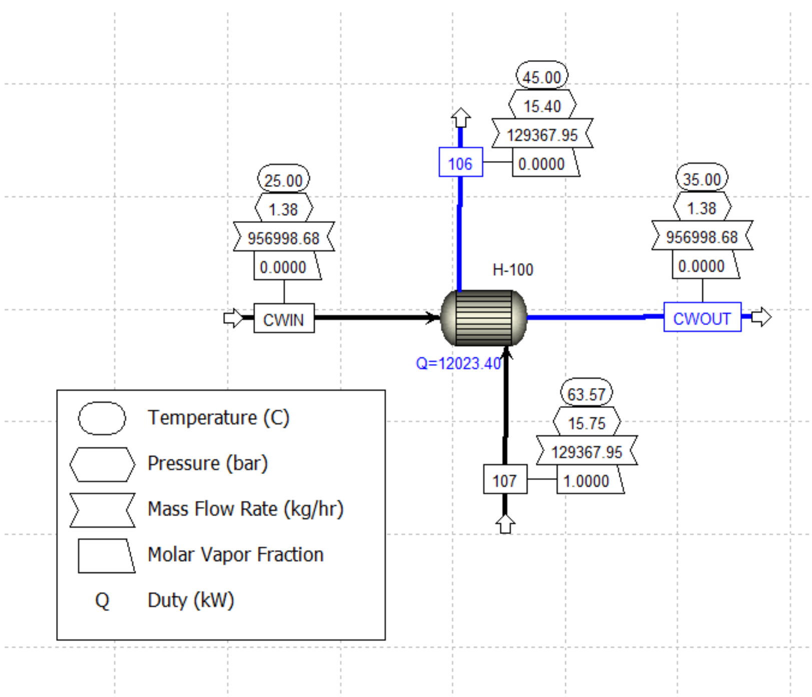


Figure 3: Condenser Hierarchy



Section #2: Heat and Weight Balance

The UIC template was used to display the stream results, and images were taken directly from Aspen Plus V14 due to the extensive amount of data, which make it hard to format it properly.

Table 1: Stream summary (100-108)

	Units	100	101	102	103	104	105	106	107	108
Description										
− MIXED Substream										
Phase		Vapor Phase	Vapor Phase	Vapor Phase	Vapor Phase	Vapor Phase	Vapor Phase	Vapor Phase	Vapor Phase	Liquid Phase
Temperature	C	-41.4765	-11.9491	-18.2198	8.23688	2.84991	24.5385	19.2781	63.569	44.9998
Pressure	bar	1.04748	2.20892	2.13997	4.00886	3.93991	6.40875	6.3398	15.748	15.4033
Molar Vapor Fraction		1	1	1	1	1	1	1	1	0
+ Mass Flows	**kg/hr**	**24276.7**	**24276.7**	**51866.3**	**51866.3**	**83342.6**	**83342.6**	**129368**	**129368**	**129368**
+ Mass Fractions										
+ Mole Flows	**kmol/hr**	**550.536**	**550.536**	**1176.2**	**1176.2**	**1890.01**	**1890.01**	**2933.78**	**2933.78**	**2933.78**
+ Mole Fractions										
Molar Enthalpy	kJ/kmol	-109401	-107596	-108019	-106406	-106792	-105495	-105895	-103446	-118199
Molar Entropy	kJ/kmol-K	-287.242	-285.846	-287.236	-286.081	-287.333	-286.457	-287.73	-286.263	-332.347
Enthalpy Flow	kW	-16730.3	-16454.3	-35292.2	-34765.1	-56065.8	-55385.3	-86298.3	-84301.9	-96325.3
Average MW		44.0965	44.0965	44.0965	44.0965	44.0965	44.0965	44.096	44.096	44.096
Molar Density	kmol/cum	0.0563098	0.107484	0.1069	0.186566	0.187575	0.292415	0.296151	0.713867	10.3913
Heat capacity, mixture	kJ/kmol-K	65.1222	69.5395	68.7203	73.8317	73.0756	78.2861	77.5869	93.1035	141.65
Availability (flow), mixture	kW	-3633.5	-3421.14	-7311.93	-6897.35	-11089.8	-10546.4	-16387.4	-14747.4	-15573.7
Exergy flow rate	kW	107.779	319.751	681.178	1095.06	1754.29	2296.83	3551.32	5189.16	4432.31

Table32: Stream summary (109-116)

	Units	109	110	111	112	113	114	115	116	117
Description										
− MIXED Substream										
Phase				Vapor Phase	Liquid Phase			Vapor Phase	Liquid Phase	
Temperature	C	10.2522	9.87124	9.87124	9.87124	-5.37282	-5.91604	-5.91604	-5.91604	-22.8028
Pressure	bar	6.40875	6.3398	6.3398	6.3398	4.00886	3.93991	3.93991	3.93991	2.20892
Molar Vapor Fraction		0.273874	0.355778	1	0	0.101082	0.377675	1	0	0.100399
+ Mass Flows	**kg/hr**	**129368**	**129368**	**46025.3**	**83342.6**	**83342.6**	**83342.6**	**31476.4**	**51866.3**	**51866.3**
+ Mass Fractions										
+ Mole Flows	**kmol/hr**	**2933.78**	**2933.78**	**1043.77**	**1890.01**	**1890.01**	**1890.01**	**713.808**	**1176.2**	**1176.2**
+ Mole Fractions										
Molar Enthalpy	kJ/kmol	-118199	-116926	-106619	-122618	-122618	-117983	-107427	-124389	-124389
Molar Entropy	kJ/kmol-K	-331.185	-326.663	-290.242	-346.778	-346.534	-329.173	-289.673	-353.145	-352.861
Enthalpy Flow	kW	-96325.3	-95287.8	-30912.9	-64374.8	-64374.8	-61941.3	-21300.7	-40640.6	-40640.6
Average MW		44.096	44.096	44.0951	44.0965	44.0965	44.0965	44.0964	44.0965	44.0965
Molar Density	kmol/cum	1.06956	0.832594	0.310475	11.6811	1.717	0.504318	0.195508	12.1765	1.04171
Heat capacity, mixture	kJ/kmol-K	105.096	101.729	76.6022	115.606	105.24	94.8889	72.0497	108.749	99.6189
Availability (flow), mixture	kW	-15856	-15917	-5823.09	-10093.9	-10132.2	-10416.1	-4176.06	-6240.01	-6267.63
Exergy flow rate	kW	4148.23	4080.39	1272.46	2807.9	2769.45	2468.67	675.736	1792.93	1765.15

Table 3: Stream summary (118-204)

	Units	118	119	120	121	200	201	202	203	204
Description										
− MIXED Substream										
Phase			Vapor Phase	Liquid Phase		Vapor Phase	Vapor Phase	Vapor Phase	Vapor Phase	Vapor Phase
Temperature	C	-23.6629	-23.6629	-23.6629	-40	25	17.2522	-0.872818	-19.3028	-38
Pressure	bar	2.13997	2.13997	2.13997	1.11643	30	29.9311	29.8621	29.7932	29.7242
Molar Vapor Fraction		0.531936	1	0	0.0886302	1	1	1	1	1
+ Mass Flows	**kg/hr**	**51866.3**	**27589.5**	**24276.7**	**24276.7**	**199747**	**199747**	**199747**	**199747**	**199747**
+ Mass Fractions										
+ Mole Flows	**kmol/hr**	**1176.2**	**625.663**	**550.536**	**550.536**	**12450.9**	**12450.9**	**12450.9**	**12450.9**	**12450.9**
+ Mole Fractions										
Molar Enthalpy	kJ/kmol	-116759	-108392	-126269	-126269	-75066.3	-75366.3	-76069.9	-76790.7	-77536.5
Molar Entropy	kJ/kmol-K	-322.252	-288.713	-360.368	-360.099	-110.059	-111.061	-113.545	-116.269	-119.304
Enthalpy Flow	kW	-38147.8	-18837.9	-19309.8	-19309.8	-259623	-260661	-263094	-265587	-268166
Average MW		44.0965	44.0965	44.0965	44.0965	16.0428	16.0428	16.0428	16.0428	16.0428
Molar Density	kmol/cum	0.204537	0.109628	12.6932	0.643954	1.29171	1.33151	1.44285	1.5827	1.76547
Heat capacity, mixture	kJ/kmol-K	84.3803	68.113	102.867	95.7664	38.9101	38.8669	38.9821	39.49	40.6264
Availability (flow), mixture	kW	-6756.52	-3877.69	-2878.83	-2891.1	-146133	-146138	-146009	-145694	-145143
Exergy flow rate	kW	1257.75	374.608	883.142	870.797	28699.2	28700.9	28844.9	29178.3	29748.4
<add properties>										

Table 4: Condenser Hierarchy stream summary

	Units	106	107	CWIN	CWOUT	
Description						
− MIXED Substream						
Phase		Liquid Phase	Vapor Phase	Liquid Phase	Liquid Phase	
Temperature	C	44.9998	63.569	25	34.9997	
Pressure	bar	15.4033	15.748	1.37895	1.37895	
Molar Vapor Fraction		0	1	0	0	
+ Mass Flows	**kg/hr**	**129368**	**129368**	**956999**	**956999**	
+ Mass Fractions						
+ Mole Flows	**kmol/hr**	**2933.78**	**2933.78**	**53121.5**	**53121.5**	
+ Mole Fractions						
Molar Enthalpy	kJ/kmol	-118199	-103446	-287741	-286926	
Molar Entropy	kJ/kmol-K	-332.347	-286.263	-167.974	-165.286	
Enthalpy Flow	kW	-96325.3	-84301.9	-4.24589e+06	-4.23387e+06	
Average MW		44.096	44.096	18.0153	18.0153	
Molar Density	kmol/cum	10.3913	0.713867	55.173	54.6346	
Heat capacity, mixture	kJ/kmol-K	141.65	93.1035	81.5008	81.4772	
Availability (flow), mixture	kW	-15573.7	-14747.4	-3.50689e+06	-3.5067e+06	
Exergy flow rate	kW	4432.31	5189.16	18.3285	142.171	
<add properties>						

Table 5: Heat Stream summary

	Q0	Q1	Q2	Q3	
QCALC kW	-2579.50514	2492.85315	2433.51222	1037.50657	
TBEGIN C	-41.4764714	-0.872817543	17.2521696	25	
TEND C	-40.000004	-19.3028476	-0.872817543	17.2521696	

Tables 1 through 3 show the stream summaries in the main flowsheet, and Table 4 displays the hierarchy stream summary for the condenser, where we used cooling water and fed it as a stream rather than a utility. The other hierarchies, like the evaporators, are not explicitly included, as they contain the same results in the main flowsheet. Table 5 displays the heat summary of the heat transfer streams used for the evaporators.

<u>Section #3:</u> Solution Logic and Description

The refrigeration loop is a four-stage propane cycle that precools/condenses methane across four evaporators (E-100 A/B, E-101 A/B, E-102 A/B, E-103 A/B) with interstage flash drums and throttling valves, followed by a condenser. Compression is in four stages with intercooling/mixing, and a condenser removes the total heat load before recycle.

The Aspen model uses explicit unit models on the main flowsheet and five hierarchies for exchanger sizing/costing: one hierarchy per evaporator (Stages 1–4) and one for the condenser. In each evaporator hierarchy the exchanger spec is (hot outlet − cold inlet) temperature difference; in the condenser hierarchy the spec is hot-side outlet vapor fraction = 0 (complete condensation). The inlets to the parallel heaters (evaporators) are transferred to their respective hierarchies via Transfer blocks so the hierarchy exchangers size under the same conditions, and the propane inlet to the condenser is likewise transferred to the condenser hierarchy; cooling-water inlet data are entered per the assignment. Because a CW mass-flow was not provided, a Design Spec sets CW outlet = 35 °C (25 °C inlet with 10 °C rise) by adjusting CW inlet mass-flow. The last valve pressure (VLV-100, leftmost) is fixed from a purity-based saturation calculation of propane at −40 °C (matching the Stage-4 evaporator at −38 °C with ΔTmin = 2 °C). A second Design Spec enforces methane outlet = 38 °C by adjusting the propane flow in the tear stream. Thus, there are two design specs and five exchanger hierarchies tied to the main flowsheet via Transfers.

Calculator Blocks

1. C-RES — performance metrics

This calculator imports the mechanical power of each compressor (W_i)and the duty of each evaporator ($Q_{e,j}$), and computes:

$$W_{\text{tot}} = \sum_i W_i\,,\ Q_{\text{evap}} = \sum_j Q_{e,j}\,,\ \text{COP} = \frac{Q_{\text{evap}}}{W_{\text{tot}}}.$$

It exports COP, total power, and any intermediate sums for reporting.

2. **C-SIM — ΔT_{min} and pressure propagation**

C-SIM applies the ΔT_{min} constraints and synchronizes pressures. It imports the cold-inlet temperatures to the first three evaporators (propane inlet to the top heater in each pair) and sets the hot-outlet temperature of the corresponding bottom exchangers (methane side) to $T_{\text{cold,in}} + \Delta T_{\text{min}}$. After manual adjusting the ΔT_{min} values of each stage, the values Stage 1: 7 °C, Stage 2: 4.5 °C, Stage 3: 3.5 °C yielded lower costs than the problem's suggested values; these same ΔT targets are also manually set inside the evaporator hierarchies' "hot-outlet – cold-inlet temperature difference" specs. C-SIM also exports imported valve discharge pressures to the compressors as for staging consistency (all except the fourth-stage compressor, which is not overwritten, and VLV-100, which does not transfer a pressure).

3. **C-COSTS — equipment and utility economics**

C-COSTS aggregates installed, capital (annualized), and operating costs by equipment and totals. Inputs:

- Exchanger areas from all five hierarchies → exchanger purchase cost via the provided area-based correlation; then installed/annualized capital (ACC) using the Lang factor, escalation, and ACCR from the problem statement.
- Compressor powers → compressor costs per the given compressor correlation.
- Condenser duty → cooling-water OPEX using the stated unit price (in $G/J, then convert).
- Valve inlet mass-flows → valve CAPEX via the valve cost correlation.
- Flash drum weights → drum CAPEX (weights taken from one economic run and assumed constant).
- Electricity OPEX via Aspen utility variable for Cost-Rate.

C-COSTS reports per-equipment Installed and ACC, CW and electricity costs, and totals (Installed, ACC, Total Annualized Cost). *(Installed ≅ ACC without the ACCR term)*

Optimizer

An Optimization block was used to minimize total compressor work. Decision variables were:

1. Valve pressures for VLV-101, VLV-102, VLV-103 (the three valves that transfer pressure), varied within reasonable bounds such that pressures decrease from right→left across valves and increase through compressor stages;
2. Compressor pressure ratios for Stages 1–3 (compressors modeled in terms of pressure ratio rather than absolute pressure).

All other constraints from the design specs (e.g., CW outlet 35 °C; methane outlet 38 °C) and ΔT_{min} targets remain active, ensuring feasible temperature approaches and stage matching while the optimizer searches for the minimum W_{tot}. Note that running this optimizer while only varying the pressure of the valves without varying pressure ratios results in a much higher price.

Section #4: Miscellaneous results

Figure 3: Key Performance Indicators

KPIs | Detailed Cost | Optimal Pressure

Name	Units	Value
COP	UNITLESS	2.33223
Total Power	KW	3663.18
Total Annualized Cost	MM$/OPER-YEAR	7.80374

Figure 3 displays the custom table of key performance indicators (KPIs) where our optimized COP (coefficient of performance) is 2.33233 our total power is 3663.18 KW, and the total annualized cost is 7.80374 MM$/operating year.

Figure 4: Detailed Cost

KPIs | Detailed Cost | Optimal Pressure

Name	Units	Value
Compressors (4)	$/OPER-Y...	9.56378e+06
Heat Exchangers (...	$/OPER-Y...	501167
Valves (4)	$/OPER-Y...	5.49855e+06
Drums (3)	$/OPER-Y...	6.04851e+06
Total Installed Cost	$/OPER-Y...	303494
Electricity	$/OPER-Y...	157131
Cooling Water	$/OPER-Y...	520284

Figure 4 displays the installed cost of the 4 compressors, 5 heat exchangers, 4 valves, 3 drums, and the Total installed cost, as well as the cost of each utility.

Figure 5: Optimal Pressures

KPIs | Detailed Cost | Optimal Pressure

Name	Units	Value
K101 Discharge	*bar*	*2.20892*
K102 Discharge	*bar*	*4.00886*
K103 Discharge	*bar*	*6.40875*
K104 Discharge	*bar*	*15.748*

Figure 5 Displays the optimal discharge pressure of each compressor, which showed a pressure ratio of about 1.9. Note that there might be slight differences in pressure ratio between compressors, since the system was directly modelled in terms of pressure ratio.

Section 5: References

All information used to solve this problem was derived from the materials provided in the problem description or from lessons taught by Dr. Michael Caracotsios. No external sources or research articles were consulted, however, the cost of equipment used information from Gavin's textbook. Aspen Plus V14 was used to perform the simulation and costing calculations, while ChatGPT was utilized to assist with navigating Aspen and refining the clarity and formatting of this report.

References

Aspen Technology, Inc. (2024). *Aspen Plus V14* [Computer software]. AspenTech. https://www.aspentech.com/

OpenAI. (2025). *ChatGPT (GPT-5)* [Large language model]. OpenAI. https://chat.openai.com/

Towler, Gavin, and Ray Sinnott. *Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design*. 3rd ed. Butterworth-Heinemann, 2021.